

IX.

Consider the control system shown in figure 4 in which a proportional compensator is employed. A specification on the control system is that the steady state error must be less than 2% for constant inputs.

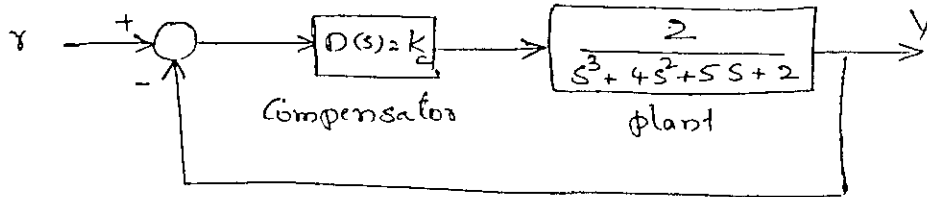


Fig. 4

- (i) Find K_c that satisfies this specification.
- (ii) If the steady-state criterion cannot be met with a proportional compensator, use a dynamic compensator $D(s) = 3 + \frac{K_1}{s}$. Find the range of K_1 that satisfies the requirement on steady-state error.

OR

X.

Explain what is meant by a PD control and show that is equivalent to adding a simple zero, to the open loop transfer function. Critically analyse the effect of this controller on the system specifications such as velocity error constant, damping ratio and undamped natural frequency of oscillation.

**B.Tech. Degree VI Semester (Supplementary) Examination
in Mechanical Engineering (CAD/CAM), December 2002**

ME 605 CONTROL SYSTEMS

(1995 Admissions)

Time: 3 Hours

Maximum Marks: 100

(All questions carry EQUAL marks)

I.

- (a) Discuss the merits and limitations of closed loop systems.
- (b) Figure 1 shows a machine of mass M mounted on a foundation using a spring of stiffness K with a damper of coefficient B in parallel. The machine generates a disturbing force $F(t) = A \sin \omega t$. Find the force transmitted to the foundation at steady-state.

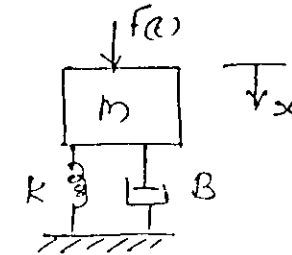


Fig. 1

OR

II.

Convert the block diagram of figure 2 to a signal flow graph, and there from obtain the input-output transfer function using Mason's gain rule.

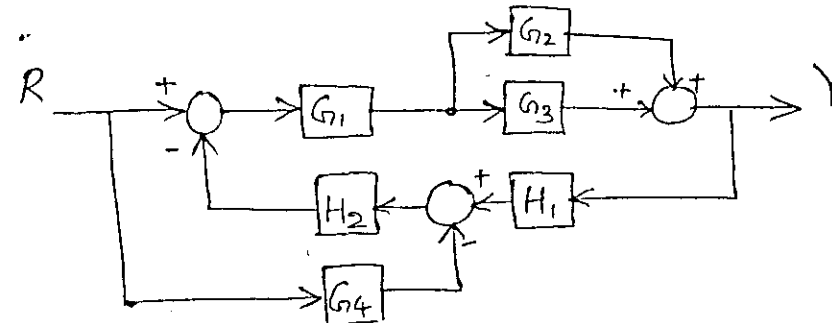


Fig. 2

(Turn over)

III. The feedback configuration of figure 3 is used to position a device in response to an input. The motor selected for this application has a gain constant $K_m = 0.5$ and a time constant $\tau_m = 0.1$. for an amplifier gain $K_A = 20$, determine the following.

- Undamped natural frequency, damping ratio, and damped natural frequency.
- Rise time, peak time, peak over shoot, and settling time.
- Position, velocity and acceleration error constants K_p , K_v and K_a .
- Steady-state error resulting from the application of (1) unit-step unit (2) unit-ramp input (3) unit-parabolic input.

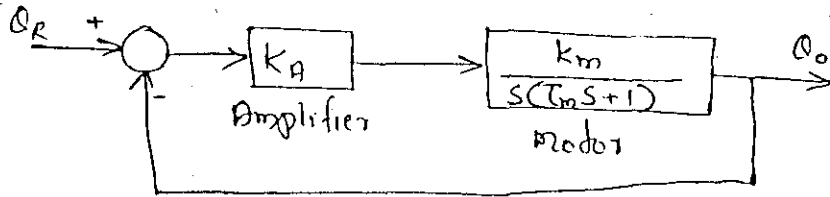


Fig. 3

OR

IV. (a) The system response when subjected to a unit step input is given by

$$C(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$$

- Obtain the closed loop transfer function.
- Undamped natural frequency.
- Damping ratio of the system.

(b) The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{K}{(s+1)}$$

The input to the system is given by $r(t) = \sin \omega t$. Compute at least four dynamic error coefficients and discuss the dynamic steady state error.

V. (a) Obtain the roots of the given characteristic polynomial on the basis of the Routh array.

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

(b) Explain the working of a hydraulic linear actuator.

OR

VI. A unity feedback control system has an open-loop transfer function

$$G(s) = \frac{K}{s(s+4)(s^2+8s+32)}; \quad K \geq 0$$

Make a rough sketch of the root locus plot of the system. Locate a point on the locus that gives dominant closed-loop poles with $\xi = 0.707$. Determine the value of K at this point. Corresponding to this value K , roughly locate the other two closed-loop poles and comment upon the dominance condition.

VII. A unity feedback system has an open-loop transfer function

$$G(s) = \frac{100}{s(1+0.1s)(1+0.2s)}$$

Obtain the Bode plots and hence comment on the stability of the system.

OR

VIII. Explain how system with transportation lag can be analysed using (i) Root locus (ii) Polar plot.